

# Managing Operational Risk in Data Operations

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## **Business Context**

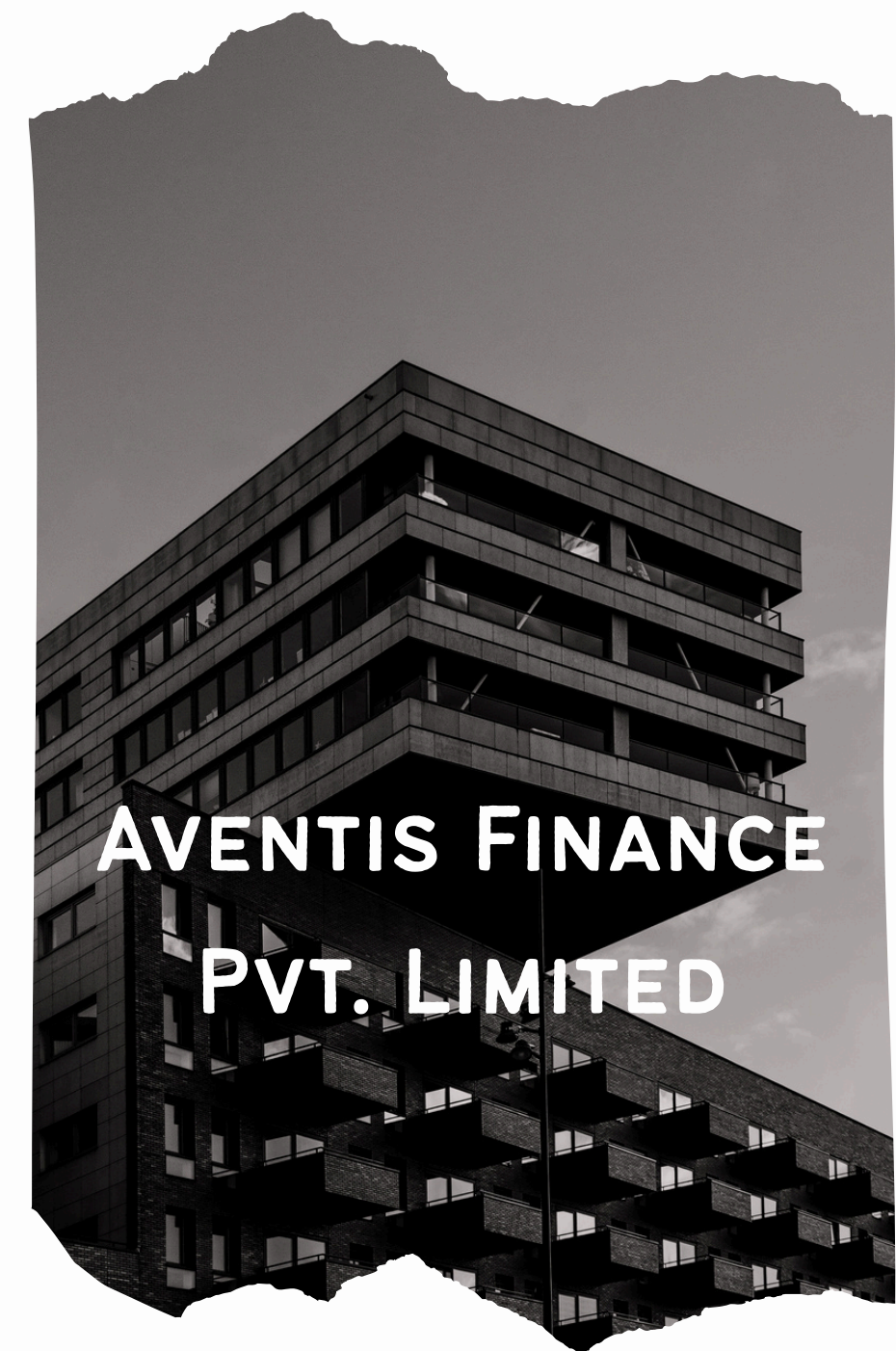
- Processes 2,025 client data feeds daily into Datahub
- Data drives real-time credit decisions via APIs
- Regulatory & SLA mandate: All uploads within T+3 days
- Penalty exposure: ₹100 per file per day

## **Core Challenge**

High-volume, high-regulation data operations with escalating financial and reputational risk

## **Streamlit Link**

<https://aventis-risk-dashboard-jxsfqopkdyfg5hhjygdpy.streamlit.app/>



# Why Proactive Risk Management Is Critical

## ***Operational Pain Points***



Late client submissions (T+1 / T+2)



Schema & format changes without notice



Manual dependency during volume spikes



System contention & silent failures

## ***Business Impact***



Regulatory non-compliance



SLA breaches & financial penalties



Reduced client trust



Reputational damage

## ***Problem Statement***

How can Aventis ensure T+3 compliance at scale while reducing operational risk and manual dependency?

# Quantified Risk Prioritization (FMEA)

## The Formula

**Severity (S):** Impact on the T+3 deadline. (10 = Regulatory Breach, 1 = Cosmetic Issue)

**Occurrence (O):** Frequency of the failure. (10 = Daily/High Volume, 1 = Once a Year)

**Detection (D):** Ability to catch the error before it causes a breach. (10 = Hidden/Client Reports it, 1 = Auto-Alert)

$$RPN = S \times O \times D$$

| Risk Dimension     | Key Risk                  | RPN  | Impact                           |
|--------------------|---------------------------|------|----------------------------------|
| Data               | Schema Drift              | 576  | Pipeline failures, hidden delays |
| People             | Manual Exception Handling | 336  | Backlogs, SLA miss               |
| Process / External | Late Client Submissions   | 108  | Compressed processing window     |
| Technology         | Resource Contention       | High | Silent queuing, downtime         |
| Regulatory         | SLA Breach                | High | Penalties & audit risk           |

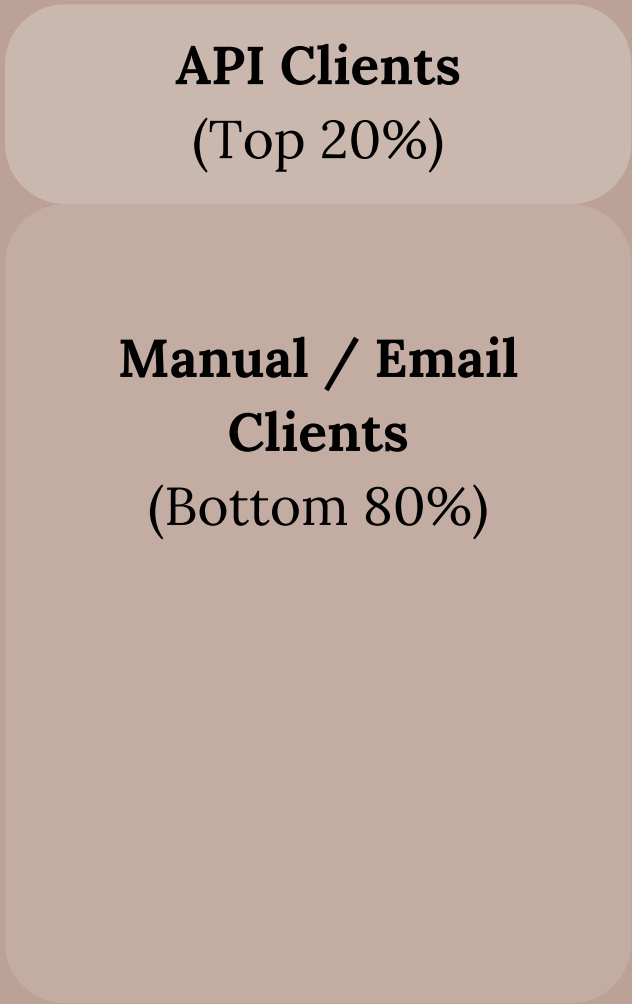
## Key Insight

Schema Drift, though less visible, is the highest systemic threat

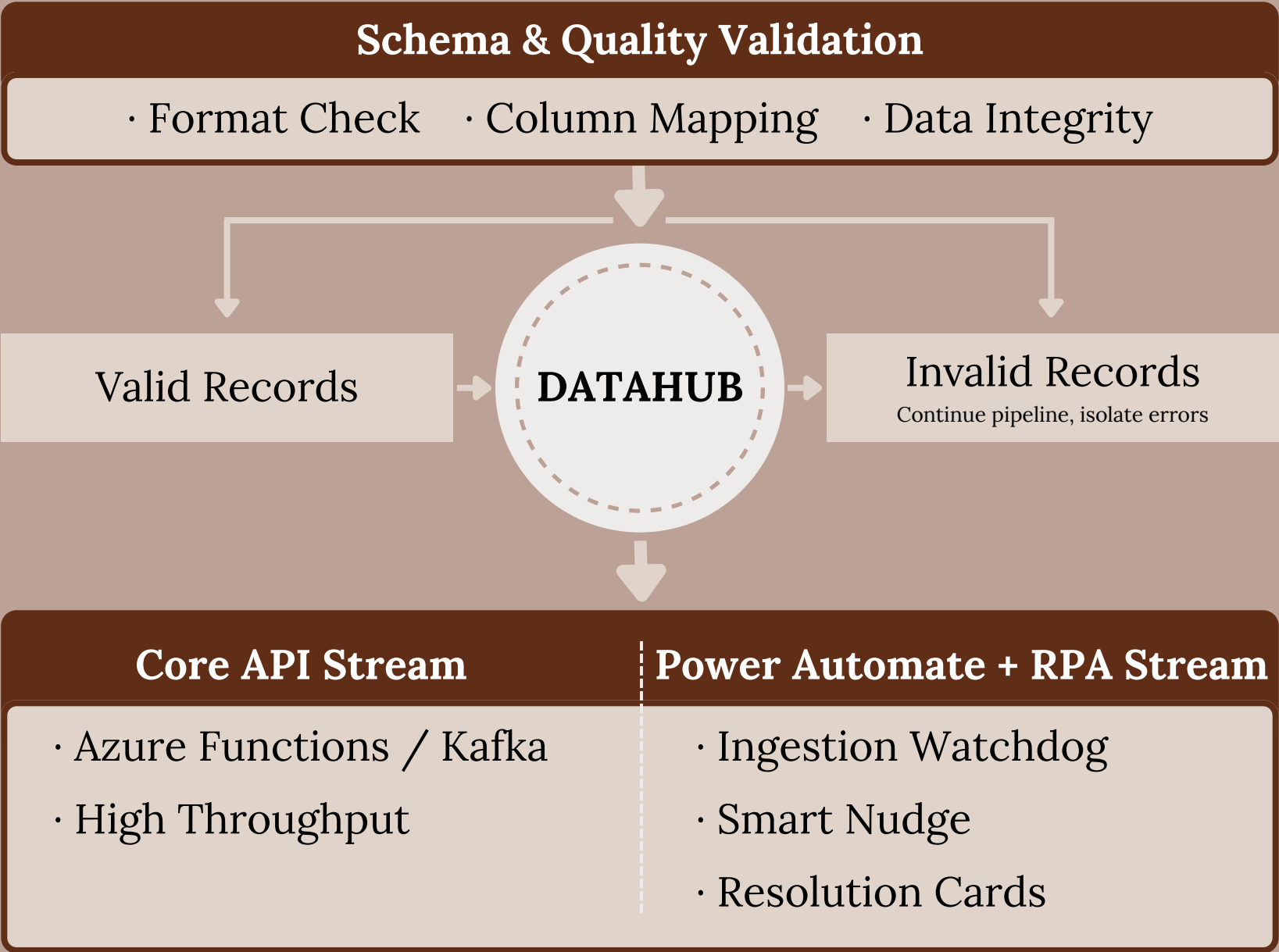
# Strategic Mitigation Architecture

Shift from "Stop-on-Error" to "Continue-and-Repair" ensures resilience at scale

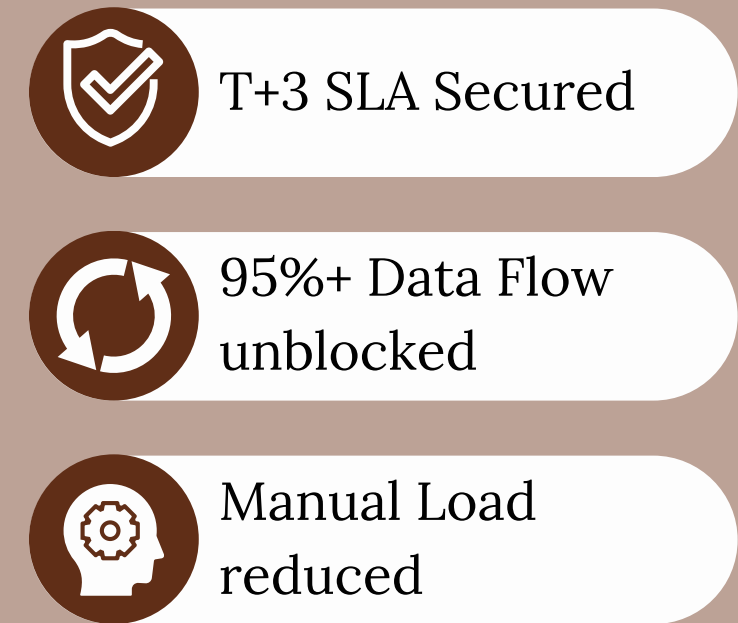
## Client Data Sources



## Hybrid Resilience Architecture



## Business Outcome



# Automation & Process Controls

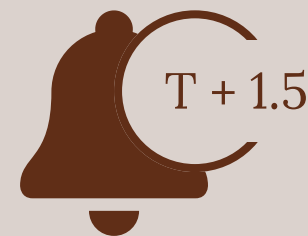
## *Ingestion Watchdog*



- Auto file pickup (email/shared drive)
- Virus scan & naming standardization
- Zero manual intake effort

Eliminates intake delays

## *Smart Nudge*



- Tracks missing submissions
- Auto-reminders to late clients
- CRM visibility for follow-up

Reduces late submissions

## *Adaptive Resolution Cards*



- Error details pushed to Ops
- One-click fix actions
- Faster MTTR

Cuts manual resolution time

**Automation replaces email chaos with controlled, SLA-safe workflows**



# Governance, KRIs & Real-Time Control



Overall SLA Status: On Track



Files Pending: 148 / 2,025



Time Buffer Remaining: 6.2 hrs

## Ops Risk Command Center

### Leading Risk Signals



Late Submissions

% files not received by T+1

Threshold: >10%



Pipeline Latency

Avg ingestion time

Threshold: >15 mins



Data Quality Index

Clean records %

Threshold: <95%



SLA Breach Risk

Buffer remaining

Threshold: <4 hrs



Manual Overload

MTTR per ticket

Threshold: >4 hrs

### Automated Escalation Model



KRI Breach Detected



Role-Based Escalation

- ✓ Ops Lead
- ✓ DevOps
- ✓ CRM
- ✓ CRO (for regulatory risk)



Predefined Playbook Activated

### Operational Metrics

Operational  
Metrics



Regulatory  
Assurance

Operational  
Signal (KRI)

Regulatory  
Outcome

Queue Depth

T+3  
Compliance

Data Quality  
Index

Reporting  
Accuracy

SLA Buffer

Audit  
Readiness

Leading indicators enable intervention hours before SLA breach

# Strategic Impact & Conclusion



## Risk State



Firefighting at T+3



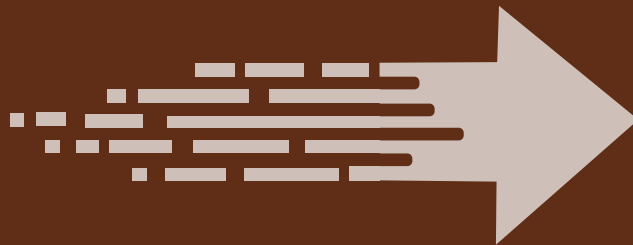
Manual exception overload



Hidden schema failures



Penalty-driven risk



## Resilient State



T+3 SLA by design



Real-time control



Scalable growth



Near-Zero Penalties

## Risk-Orchestrated Model

FMEA-based  
prioritization

Quarantine  
Pattern

Hybrid  
Automation

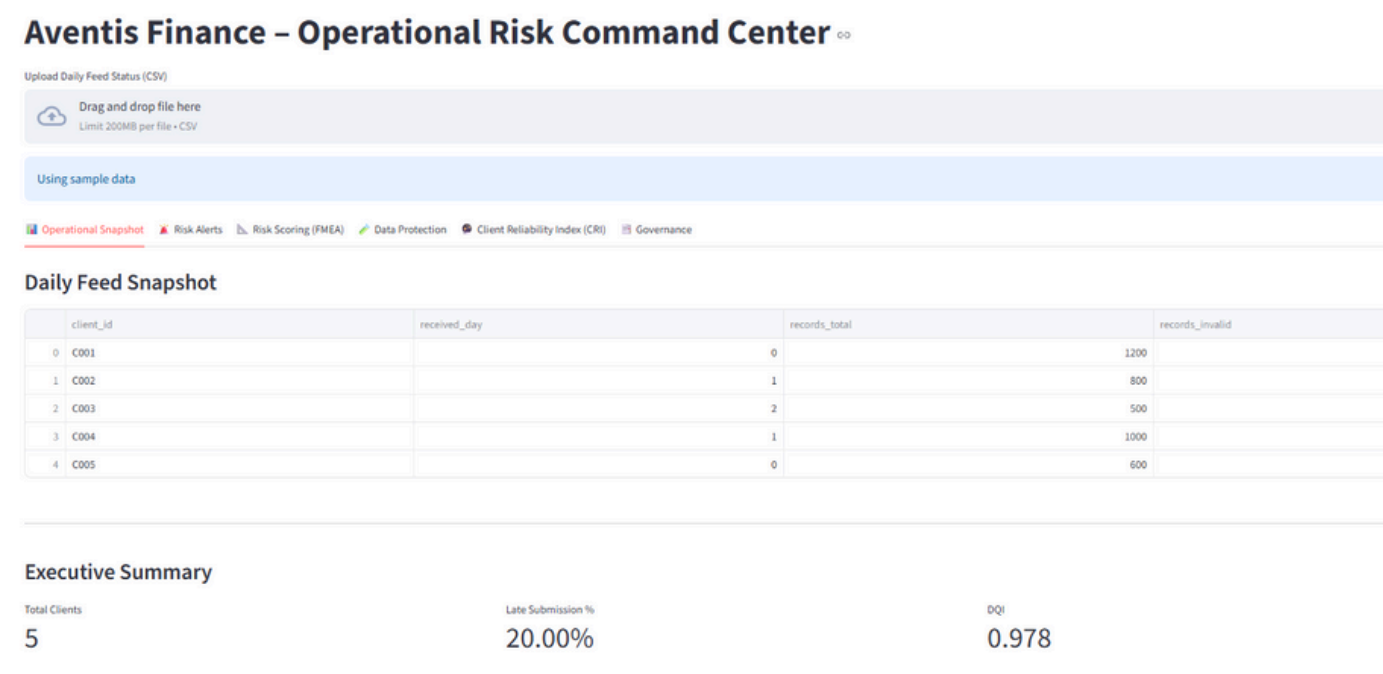
KRIs & Predictive  
Alerts

Data operations engineered for compliance

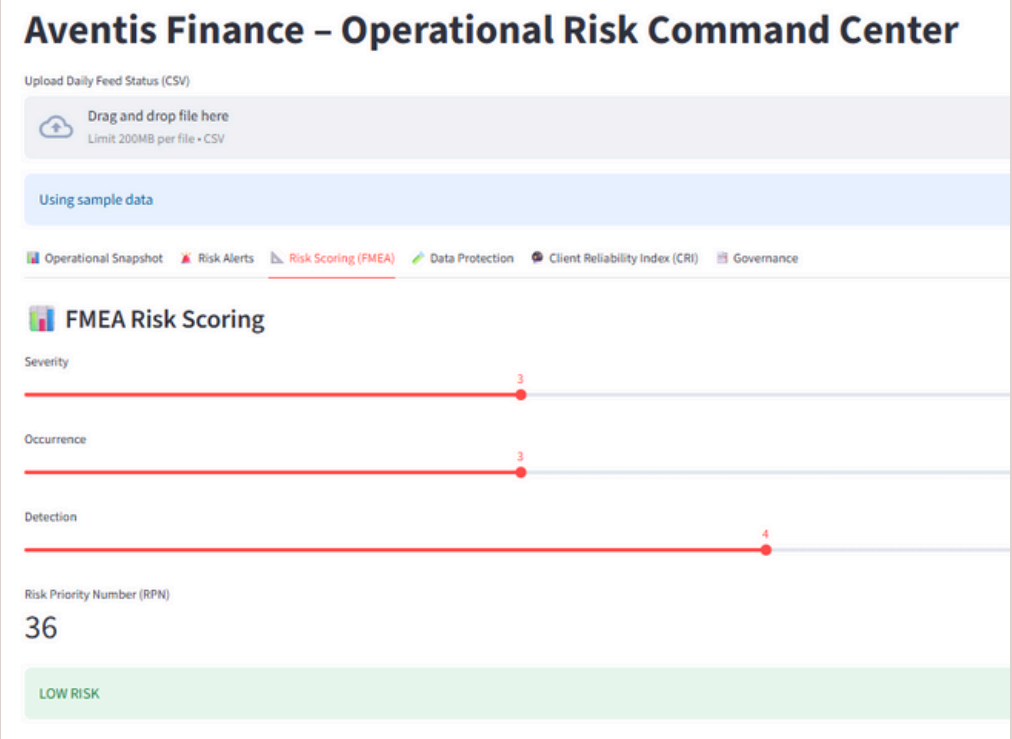
## Business Impact Delivered

- Regulatory Compliance (BCBS 239 aligned)
- Client Trust & SLA Credibility
- Cost & Penalty Reduction
- License to Scale

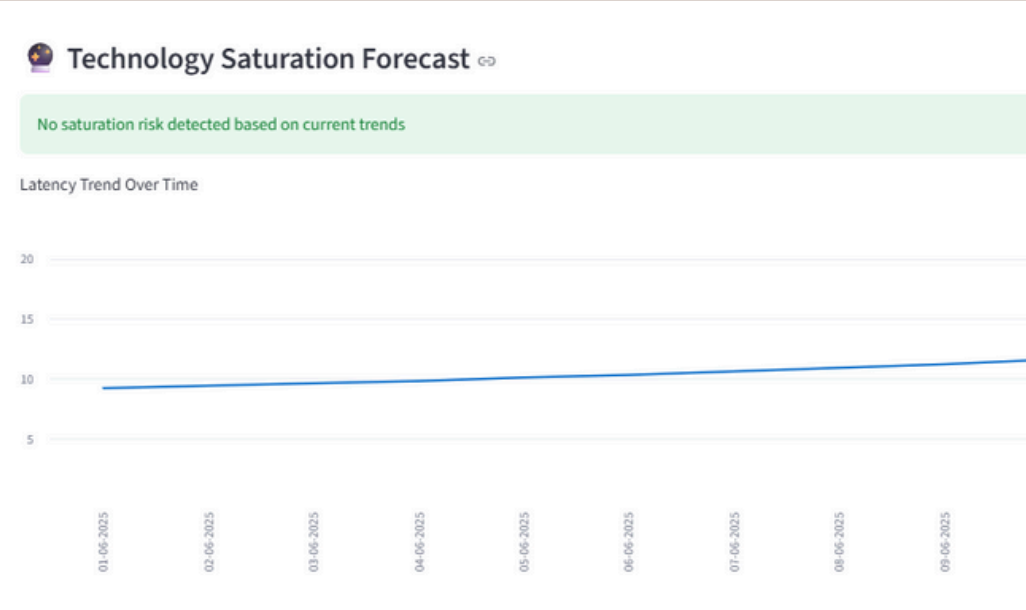
# Snap from Model Deployment( Streamlit)



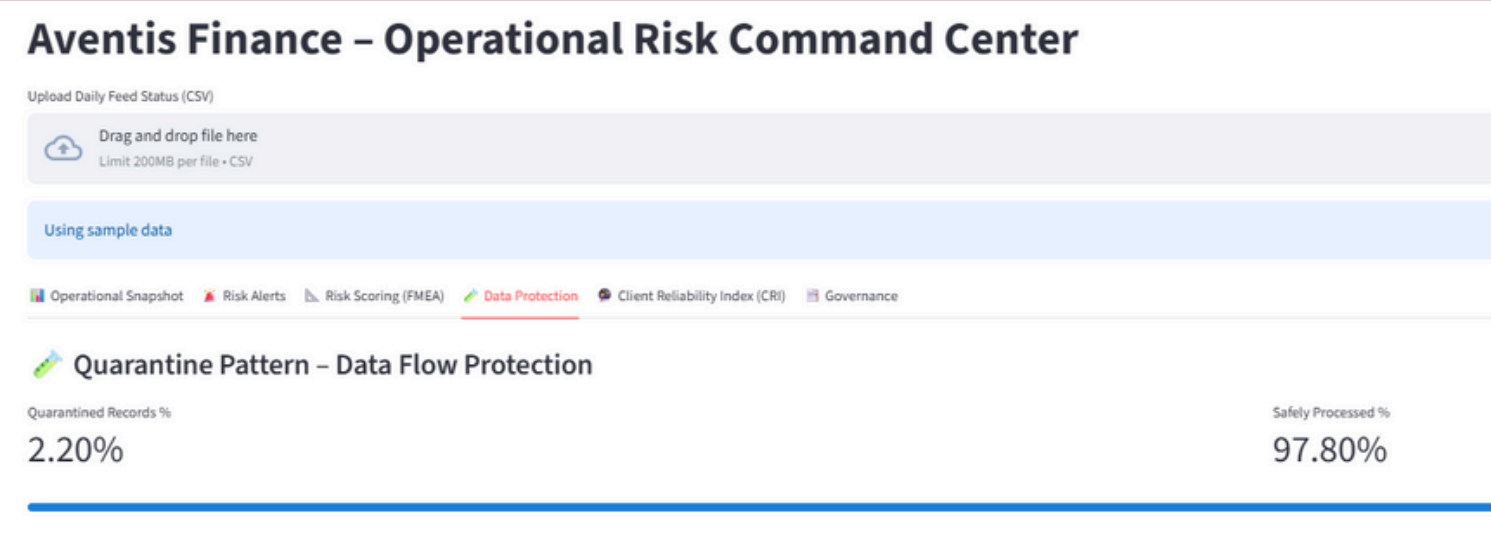
Operational Snap



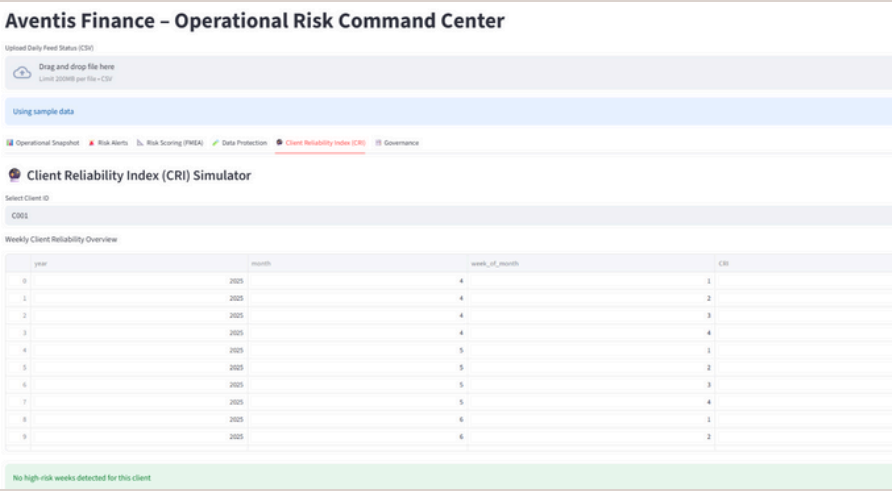
Risk Scoring(FMEA)



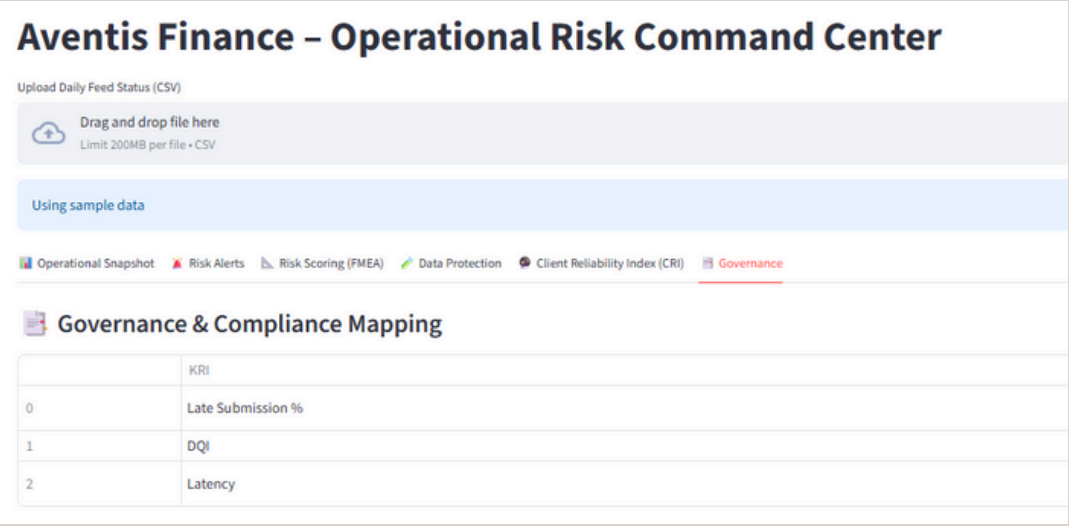
Technology Saturation Forecast



Data Protection



Client Reliability Index



Governance